



CORE COMPUTER SCIENCE

Computer Networks

Placement Preparation Handbook

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Computer Networks Roadmap

PLACEMENT PREPARATION GUIDE

How to use this Handbook:

Every single page covers exactly one critical core topic. Spend 2 minutes reading the **Definition** and **Visual Diagram** on the left, memorize the **Model Answer** and **Buzzwords** on the right, and review the **Interviewer Trap** to avoid falling into common placement traps.

Three-Phase Learning Plan

PHASE 1

Basics & Models

Master OSI & TCP/IP architectures, IP addressing schemas, and MAC mapping logic. (Topics 1 – 6)

PHASE 2

Transport & App Core

Deep-dive into TCP state machines, sliding windows, congestion control algorithms, DNS, and HTTP headers. (Topics 7 – 15)

PHASE 3

Routing & Security

Learn routing policies (BGP vs OSPF), NAT logic, stateful firewalls, VPN tunneling, and DDoS mitigations. (Topics 16 – 24)

Target Completion: 2 Hours Core Study & 30 Mins Self-Recall

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NETWORK ARCHITECTURE & MODELS

Yield: ★★★★★

01 – The OSI Reference Model

The 7-layer theoretical framework for network communication.

DEFINITION & KEY CONCEPTS

Standardizes network communication into 7 logical layers. Separates concerns: software/user space (L5–7) vs transport/routing (L3–4) vs hardware/cables (L1–2).

L#	Layer Name	PDU	Devices / Protocols
7	Application	Data	HTTP, DNS, SMTP
6	Presentation	Data	SSL/TLS, ASCII, JPEG
5	Session	Data	Sockets, RPC, NetBIOS
4	Transport	Segment	TCP, UDP, Ports
3	Network	Packet	IP, ICMP, Routers
2	Data Link	Frame	Ethernet, MAC, Switches
1	Physical	Bit	Cables, Hubs, Repeaters

REAL-WORLD TROUBLESHOOTING USE-CASE

Used to isolate bugs bottom-up. If a client can't fetch data: test L1 (cable/WiFi status) → L3 (ping server IP) → L4 (telnet to port) → L7 (verify HTTP payload response).

WHERE USED IN INDUSTRY

Web developers utilize the OSI model conceptually to isolate network bugs, deciding if a connection error is a routing issue (L3) or a TLS failure (L6).

INTERVIEW DEPTH

L1: Purpose of the 7 layers and layer boundaries.
L2: Data Encapsulation formats (Segment, Packet, Frame, Bits).
L3: Comparison: Presentation (encryption/compression) vs Application layer.
L4: Physical hardware mapping (Cables at L1, Switches at L2, Routers at L3).
L5: Decapsulation sequence and error boundary checks during packet arrival.

CORE INTERVIEW QUESTION

"At which layer does a Router operate, and how does it differ from a Layer 2 Switch?"

KEYWORDS TO DROP

LOGICAL ADDRESSING

MAC FORWARDING

LAYER 3 PACKET

LAYER 2 FRAME

MODEL ANSWER

"A Router operates at Layer 3 (Network). It reads IP packets and forwards them between different subnets using routing tables. A Switch operates at Layer 2 (Data Link) and forwards Ethernet frames within the same local network using MAC address tables."

DEEP DIVE FOLLOW-UP

→ **"What is a Layer 3 Switch?"**

A hardware-optimized switch capable of routing IP packets at wire speed between VLANs, combining routing tables with MAC learning.

INTERVIEW DEPTH PROGRESSION

Level 1 (Basic): List all 7 layers and their physical order bottom-up.
Level 2 (PDU): Map layer data units (Segment, Packet, Frame, Bit).
Level 3 (Device): Explain why routers operate at L3 and switches at L2.
Level 4 (Gateway): Understand why firewalls/gateways run up to L7.

COMMON MISTAKE

Saying the internet runs on OSI. It runs on TCP/IP.

TOP FOLLOW-UPS

- Why does a gateway operate at all 7 layers?
- What is encapsulation/decapsulation?

INTERVIEWER TRAP

Saying that the OSI model is actively used on the internet. It is strictly a theoretical reference model; the internet runs on the TCP/IP model.

MEMORY TRICK

Please Do Not Throw Sausage Pizza Away (Physical, Data Link, Network, Transport, Session, Presentation, Application)

NETWORK ARCHITECTURE & MODELS

Yield: ★★★★★

02 – TCP/IP Model Suite

The practical 4-layer architecture powering the Internet.

PRACTICAL LAYER CONSOLIDATION

The Transmission Control Protocol/Internet Protocol suite collapses the 7 theoretical OSI layers into 4 highly functional layers optimized for real implementation.

Application (OSI 5,6,7)
HTTP, DNS, SSH

↓↑ Port Addressing

Transport (OSI 4)
TCP, UDP

↓↑ Logical Routing

Internet (OSI 3)
IP, ICMP

↓↑ Physical Delivery

Network Access (OSI 1,2)
Ethernet, Wi-Fi

WHY TCP/IP WON OVER OSI

OSI was designed by standard committees before protocols were fully written. TCP/IP was built by practitioners (DARPA) who implemented software first. Practice beat theory.

WHERE USED IN INDUSTRY

The modern Internet backbone and cloud environments (like AWS VPCs) operate strictly on the 4-layer TCP/IP suite to route packet streams globally.

INTERVIEW DEPTH

- L1:** 4-layer TCP/IP layout vs 7-layer OSI layout.
- L2:** Link, Internet, Transport, and Application layers.
- L3:** Header overhead across encapsulation layers.
- L4:** Multiplexing/demultiplexing using protocol numbers and port selectors.
- L5:** Kernel networking stack architecture (socket queues, ring buffers).

CORE INTERVIEW QUESTION

"How does the TCP/IP Model handle data delivery differently from OSI?"

KEYWORDS TO DROP

INTERNET LAYER

CONCISE LAYERS

HORIZONTAL ROUTING

END-TO-END RELIABILITY

MODEL ANSWER

"TCP/IP combines the Application, Presentation, and Session layers into a single Application layer, leaving data formatting and session state to the application itself. The Internet layer represents the Network layer, ensuring global routing via IP protocols."

DEEP DIVE FOLLOW-UP

→ **"Which layer handles encryption in TCP/IP?"**

Typically the Application layer (e.g., TLS handles encryption for HTTPS before data reaches TCP transport layer).

REAL INTERVIEW SCENARIO

Interviewer: "Why did the TCP/IP model win over the theoretical OSI model?"

Candidate: "OSI was designed by standard committees before software protocols were fully implemented. TCP/IP was built by DARPA engineers who wrote working code first. Practice beat theory."

COMMON MISTAKE

Confusing layer names (e.g. Network vs Internet).

TOP FOLLOW-UPS

- How does the Internet layer handle routing?
- Which layer handles encryption in TCP/IP?

INTERVIEWER TRAP

Don't confuse the TCP/IP Internet Layer name with the OSI Network Layer name, though they do the exact same job.

MEMORY TRICK

TCP/IP is OSI in a tight corset — squeezed from 7 layers down to a highly practical 4.

TRANSPORT LAYER PROTOCOLS

Yield: ★★★★★

03 – TCP vs UDP

Reliability and ordering vs raw speed and low overhead.

CORE PROTOCOL TRADE-OFFS

TCP: Connection-oriented, guaranteed in-order delivery, flow control, congestion control. Slower due to handshakes and headers.

UDP: Connectionless, unconfirmed delivery, fast, thin header (8 bytes vs 20 bytes). Broadcast/multicast support.

Feature	TCP	UDP
State	Stateful	Stateless
Header	20–60 bytes	8 bytes
Delivery	Guaranteed	Best-effort
Flow Control	Yes (Sliding Window)	No
Use-case	HTTP, SSH, Email	DNS, VoIP, Gaming

REAL-WORLD API DESIGN DECISION

Zoom uses UDP for live audio/video frames (speed matters; dropped frames are skipped) but uses TCP for chat messages and authentication (no loss tolerated).

WHERE USED IN INDUSTRY

Real-time communication software (like Zoom or Discord) uses UDP for video/voice data streaming to eliminate lag, while using TCP for reliable chat messages.

INTERVIEW DEPTH

- L1:** Connection-oriented (TCP) vs connectionless (UDP) behaviors.
- L2:** Reliability tools: ACKs, checksums, and sequence IDs in TCP.
- L3:** Header comparisons: 20–60 byte TCP header vs 8-byte UDP header.
- L4:** Use cases: gaming/streaming (UDP) vs web page/file transfer (TCP).
- L5:** User-space reliability layers (e.g. QUIC protocol) running on top of UDP.

CORE INTERVIEW QUESTION

"Why does UDP have a checksum if it is called an 'unreliable' protocol?"

KEYWORDS TO DROP

DATA INTEGRITY

SILENT CORRUPTION

ERROR DETECTION

NO RETRANSMISSION

MODEL ANSWER

"UDP provides error detection but NOT error correction. The UDP checksum identifies if bits were corrupted in transit. If corrupt, the packet is discarded. It is 'unreliable' because it won't ask for a retransmission, but it won't deliver bad data."

DEEP DIVE FOLLOW-UP

→ "What is HTTP/3's connection model?"

HTTP/3 runs over QUIC, which uses UDP to eliminate Head-of-Line blocking while implementing custom reliability and congestion control in user space.



PROTOCOL TRADE-OFF ANALYSIS

TCP (Stateful): High overhead (20B header), guarantees order, handles congestion. Best for files, web, and database connections.

UDP (Stateless): Low overhead (8B header), no delivery checks. Best for DNS, VoIP, real-time gaming, and multiplexed streams.

COMMON MISTAKE

Thinking UDP is 'broken'. It simply lacks delivery checks.

TOP FOLLOW-UPS

- Why does DNS use UDP instead of TCP?
- Can we build a reliable protocol on top of UDP?



INTERVIEWER TRAP

Never say UDP is 'bad' or 'insecure' just because it's unreliable. UDP is the default choice for high-frequency, low-latency applications.



MEMORY TRICK

TCP is a certified mail delivery with signature receipt. UDP is throwing a newspaper at a moving target.

NETWORK LAYER ADDRESSING

Yield: ★★★★★

04 – IP Addressing (IPv4 vs IPv6)

Logical identification systems for global internet routing.

FORMAT & COMPOSITION

IPv4: 32-bit address, dotted-decimal format (e.g., 192.168.1.1). Total address space ~4.3 billion (exhausted).

IPv6: 128-bit address, hexadecimal colon format (e.g., 2001:db8::ff00:42). Total space $\sim 3.4 \times 10^{38}$. Built-in IPsec.

IPv4 Address Structure

192.168.1.1 (4 Octets)
[Network Part] [Host Part]

IPv6 Address Structure

2001:0db8::8a2e:0370:7334
(8 groups of 16-bit hex values)

LOOPBACK ADDRESSES

IPv4 uses 127.0.0.1 for loopback (localhost). IPv6 uses ::1. Loopback traffic never leaves the local network interface card (NIC).

WHERE USED IN INDUSTRY

Enterprise cloud architectures route traffic using Classless Inter-Domain Routing (CIDR) blocks (e.g. /16 or /24) to partition large subnets.

INTERVIEW DEPTH

- L1:** IPv4 address classes (A, B, C) and private IP ranges.
- L2:** Classless Inter-Domain Routing (CIDR) prefix formats.
- L3:** IPv6 features (128-bit space, SLAAC, header layout differences).
- L4:** Loopback addresses (127.0.0.1) and APIPA ranges (169.254.x.x).
- L5:** Hierarchical IP routing blocks and prefix aggregation.

CORE INTERVIEW QUESTION

"What was the primary driver for IPv6, and what does it change other than address length?"

KEYWORDS TO DROP

ADDRESS EXHAUSTION

SIMPLIFIED HEADER

NO BROADCASTS

AUTOCONFIGURATION

MODEL ANSWER

"IPv6 was created to solve IPv4 address exhaustion. Key architectural upgrades include: a fixed-size simplified header for faster router processing, native SLAAC (Stateless Address Autoconfiguration), replacing Broadcasts with Multicasts, and removing IP-level checksums."

DEEP DIVE FOLLOW-UP

→ "Why are broadcasts removed in IPv6?"

Broadcasts force all devices on a local subnet to process the packet. IPv6 replaces broadcasts with targeted multicast groups to save device CPU cycles.

ADDRESS DEPTH PROGRESSION

Level 1 (Basic): Differentiate 32-bit IPv4 from 128-bit IPv6 address size.

Level 2 (Format): Format IPv4 in dotted decimal and IPv6 in hexadecimal.

Level 3 (Exhaustion): Explain how NAT/CIDR delayed IPv4 address depletion.

Level 4 (IPsec): Identify that IPsec is built natively into the IPv6 standard.

COMMON MISTAKE

Believing that textbook definitions are sufficient; always lead with real-world trade-offs.

TOP FOLLOW-UPS

- What is the purpose of a subnet mask?
- Explain private vs public IP addressing.

INTERVIEWER TRAP

Don't confuse Classful Routing (A, B, C) with classless CIDR routing. Classful routing has been obsolete since 1993.

MEMORY TRICK

IPv4 = 4 bytes (32-bit). IPv6 = 16 bytes (128-bit). Address space is large enough to assign an IP to every atom on Earth.

NETWORK LAYER ADDRESSING

Yield: ★★★★★

05 – Subnetting & CIDR

Dividing single address spaces into multiple logical subnets.

KEY EQUATIONS & MASKS

CIDR notation dictates the network prefix bits (e.g. /24 has 24 network bits, 8 host bits). Host bits determine IPs.

Usable Hosts Formula: $2^{(32 - \text{CIDR})} - 2$

CIDR	Subnet Mask	Total IPs	Usable Hosts
/24	255.255.255.0	256	254
/25	255.255.255.128	128	126
/26	255.255.255.192	64	62
/27	255.255.255.224	32	30
/30	255.255.255.252	4	2 (P2P WAN)

CLOUD VPC DESIGN

AWS VPCs are typically allocated a /16 block. Subnets inside are allocated /24 blocks, separating Web, App, and Database layers logically.

WHERE USED IN INDUSTRY

Network security teams partition corporate offices into logical subnets to isolate guest Wi-Fi networks from sensitive internal backend databases.

INTERVIEW DEPTH

- L1:** Calculating network ID, host ID, and broadcast IP.
- L2:** Subnet masks conversion to binary representation.
- L3:** Variable Length Subnet Masking (VLSM) partitioning schemas.
- L4:** Usable host range equations: $2^{(32-N)} - 2$.
- L5:** Inter-VLAN routing interfaces and route table setups.

CORE INTERVIEW QUESTION

"Why are there exactly 2 subtracted host addresses in every subnet calculation?"

KEYWORDS TO DROP

NETWORK ADDRESS

BROADCAST ADDRESS

HOST BITS ALL ZEROES

HOST BITS ALL ONES

MODEL ANSWER

"The first address (host bits all 0s) is the Network Identifier, which routers use to route to the subnet. The last address (host bits all 1s) is the Broadcast Address for sending packets to all hosts in the subnet. Neither can be assigned to a host device."

DEEP DIVE FOLLOW-UP

→ **"What is the subnet size of a /32 block?"**

Exactly 1 IP. Used for individual host loopbacks or routing tables targeting a single specific server.

REAL INTERVIEW SCENARIO

Interviewer: "If you allocate /26 subnets in your cloud VPC, how many servers can you run in each?"

Candidate: "A /26 subnet has 6 host bits, giving 64 total addresses. We subtract 2 for the network and broadcast addresses, leaving 62 usable host IPs."

COMMON MISTAKE

Believing that textbook definitions are sufficient; always lead with real-world trade-offs.

TOP FOLLOW-UPS

- How do you calculate the network address of a subnet?
- What is the size of a /32 subnet?

INTERVIEWER TRAP

Don't miscalculate a /30 subnet! A /30 has 4 total IPs, but only 2 are usable, making it perfect for connecting exactly two routers.

MEMORY TRICK

h is host bits. Subtract h from 32, double it iteratively: /30 has 2 host bits, $2^2 = 4$ total IPs.

DATA LINK LAYER & RESOLUTION

Yield: ★★★★★

06 – MAC Address & ARP

Mapping Layer 3 IP logical addresses to Layer 2 MAC physical addresses.

LOGICAL VS PHYSICAL ADDRESS

MAC (Media Access Control): 48-bit burned-in physical hardware address, unique globally (e.g., 00:0a:95:9d:68:16).

ARP (Address Resolution Protocol): Dynamic mapping helper protocol. Discovers the MAC address of an IP on the local subnet.

ARP Request (Broadcast)
"Who has IP 192.168.1.5? Tell MAC 00:0A..."



ARP Reply (Unicast)
"IP 192.168.1.5 is at MAC 00:0B:44..."

SECURITY EXPLOIT: ARP SPOOFING

Because ARP lacks authentication, an attacker can send fake ARP replies claiming their MAC address matches the default gateway IP, executing a Man-in-the-Middle (MitM) intercept.

WHERE USED IN INDUSTRY

Local area switches run Address Resolution Protocol (ARP) tables to translate dynamic IP addresses to static physical MAC card slots on local hardware.

INTERVIEW DEPTH

- L1:** MAC physical addresses vs IP logical addresses.
- L2:** ARP DORA phases: Request (broadcast) and Reply (unicast).
- L3:** ARP cache tables timeouts and local sweeps.
- L4:** ARP Spoofing attacks and dynamic ARP inspection (DAI) guards.
- L5:** Proxy ARP routing across subnet interfaces.

CORE INTERVIEW QUESTION

"How does a packet's MAC address and IP address change as it travels across different routers?"

KEYWORDS TO DROP

END-TO-END CONSTANT

HOP-BY-HOP REWRITE

SOURCE/DEST MAC

DEFAULT GATEWAY

MODEL ANSWER

"The source and destination IP addresses remain constant end-to-end (from sender to receiver). However, the source and destination MAC addresses are rewritten at every router hop as the frame is de-encapsulated and re-encapsulated for the next physical link."

DEEP DIVE FOLLOW-UP

→ "What is gratuitous ARP?"

An unprompted ARP reply broadcasted by a host to update the ARP tables of all other hosts on the network (e.g., when a backup server takes over an IP).

RESOLVE DEPTH PROGRESSION

- Level 1 (Logical):** Explain that ARP maps a Layer 3 IP to a Layer 2 MAC.
- Level 2 (Process):** Describe the ARP broadcast request and unicast reply flow.
- Level 3 (Caching):** State why ARP tables cache MAC mappings locally for speed.
- Level 4 (Spoofing):** Explain how gratuitous ARP allows ARP spoofing MitM attacks.

COMMON MISTAKE

Saying ARP is routed globally. It is local broadcast only.

TOP FOLLOW-UPS

- How does a packet's MAC address change at router hops?
- What is gratuitous ARP?

INTERVIEWER TRAP

Don't say ARP broadcasts travel across routers. ARP is local; routers block broadcasts. You can only resolve MAC addresses on your local subnet.

MEMORY TRICK

IP tells you where the package is going globally. MAC tells you who takes it to the next physical truck.

APPLICATION LAYER SERVICES

Yield: ★★★★★

07 – Domain Name System (DNS)

The distributed database translating human names to IP addresses.

HIERARCHICAL ARCHITECTURE

Uses a tree-structured database space. Resolves names recursively via Root, Top-Level Domain (TLD), and Authoritative servers.

Client → **Resolver** (ISP/1.1.1.1)
Resolver → **Root Server** (".")
Resolver → **TLD Server** (".com")
Resolver → **Authoritative Server** ("google.com")
Resolver returns IP to Client.

COMMON DNS RECORD TYPES

A: IPv4 address. **AAAA:** IPv6 address. **CNAME:** Canonical name (alias). **MX:** Mail server. **TXT:** Domain verification codes.

WHERE USED IN INDUSTRY

Content Delivery Networks (like Cloudflare) use Geolocation DNS routing to direct domain requests to the nearest edge cache server, minimizing latency.

INTERVIEW DEPTH

L1: DNS domain names hierarchy (Root, TLD, Authoritative DNS).
L2: Recursive vs Iterative query resolutions.
L3: DNS record classes (A, AAAA, CNAME, MX, TXT, NS).
L4: TTL (Time-To-Live) cache invalidation settings on clients.
L5: Anycast DNS routing and DNSSEC cryptographic validations.

CORE INTERVIEW QUESTION

"Explain the difference between Recursive and Iterative DNS queries."

KEYWORDS TO DROP

LOCAL RESOLVER

AUTHORITATIVE ANSWER

ROOT HINTS

DNS CACHING

MODEL ANSWER

"A Recursive query forces the server (resolver) to do all the work and return the final IP address or an error. In an Iterative query, the DNS server returns the best referral it has (e.g., 'I don't know the IP, but ask this TLD server')."

DEEP DIVE FOLLOW-UP

→ "What transport protocol does DNS use?"

DNS queries typically use UDP Port 53 for speed. However, zone transfers or large responses >512 bytes fallback to TCP Port 53.

REAL INTERVIEW SCENARIO

Interviewer: "Why does DNS run on UDP instead of TCP?"

Candidate: "Speed and scale. DNS requests are small, fitting in one packet. UDP avoids connection setup round-trips. But if a response exceeds 512 bytes, DNS falls back to TCP."

COMMON MISTAKE

Not knowing recursive vs iterative lookup differences.

TOP FOLLOW-UPS

- Differentiate recursive and iterative DNS queries.
- What is DNS cache poisoning?

INTERVIEWER TRAP

Don't assume DNS changes take effect instantly. TTL (Time To Live) forces resolvers to cache old records until the TTL expires.

MEMORY TRICK

Recursive = 'Get me the answer!' Iterative = 'Give me a phone book recommendation!'

APPLICATION LAYER SERVICES

Yield: ★★★★★

08 – DHCP Protocol

Dynamically allocating IP configurations to connecting hosts.

DYNAMIC ALLOCATION

Eliminates static IP configuration. Automatically leases IP address, Subnet Mask, Default Gateway, and DNS servers to client hosts.

```
[Client]                                [DHCP Server]
— DISCOVER (Broadcast) —>
<— OFFER (Unicast/Bcast) —
— REQUEST (Broadcast) —>
<— ACKNOWLEDGE (Unicast) —
```

DHCP LEASES

IPs aren't given forever. A host requests a renew halfway through the lease time (T1 threshold, typically 50% of lease duration) via unicast.

WHERE USED IN INDUSTRY

Public Wi-Fi routers dynamically assign IP configurations to thousands of customer smartphones using DHCP lease timers to recycle idle IP space.

INTERVIEW DEPTH

- L1:** DHCP DORA sequence: Discover, Offer, Request, Acknowledge.
- L2:** DHCP lease negotiation timers (T1 at 50%, T2 at 87.5%).
- L3:** DHCP Relay Agents forwarding broadcasts across subnet links.
- L4:** IP pool exhaustion and static MAC-IP reservations.
- L5:** Rogue DHCP server attacks and DHCP Snooping protection.

CORE INTERVIEW QUESTION

"Why is the DHCP Request packet sent as a broadcast if the server already offered an IP?"

KEYWORDS TO DROP

DORA PROCESS

MULTIPLE SERVERS

IMPLICIT DECLINE

PORT 67 / 68

MODEL ANSWER

"Multiple DHCP servers can hear a DISCOVER and send an OFFER. The client broadcasts the REQUEST to notify all listening servers which offer it accepted, allowing the rejected servers to release their reserved IPs back into the pool."

DEEP DIVE FOLLOW-UP

→ "What is DHCP Relay?"

A feature on a router that catches local broadcast DHCP Discover packets and forwards them as unicast packets to a DHCP server on a different network subnet.

DHCP ALLOCATION STATES

- Level 1 (DORA):** Define Discover, Offer, Request, and Acknowledge stages.
- Level 2 (Port):** Recall that DHCP uses UDP ports 67 (server) and 68 (client).
- Level 3 (Leasing):** Explain T1 (renewal at 50% time) and T2 (rebinding at 87.5% time).
- Level 4 (Relay):** Understand how DHCP relay agents route requests across subnets.

COMMON MISTAKE

Thinking IP lease is permanent. IPs are leased for temporary sessions.

TOP FOLLOW-UPS

- Describe the DHCP DORA process steps.
- What is DHCP relaying?

INTERVIEWER TRAP

Don't confuse APIPA (169.254.x.x link-local IP assigned when DHCP fails) with a valid DHCP lease assignment.

MEMORY TRICK

Remember the DORA acronym: Discover, Offer, Request, Acknowledge.

APPLICATION LAYER PROTOCOLS

Yield: ★★★★★

09 – HTTP Evolution

The protocol of the web: comparative analysis from HTTP/1.1 to HTTP/3.

HISTORY & EVOLUTION

HTTP/1.1: Added persistent connections (Keep-Alive) but suffered from Head-of-Line (HoL) blocking on the TCP level.

HTTP/2: Multiplexed streams over a single TCP connection, binary framing, HPACK header compression, Server Push.

HTTP/3: Replaces TCP with QUIC (over UDP) to eliminate TCP-level HoL blocking during packet loss.

CORE INTERVIEW QUESTION

"What is Head-of-Line (HoL) blocking, and how does HTTP/3 solve it?"

KEYWORDS TO DROP

TCP BUFFER BLOCK

QUIC STREAMS

UDP TRANSPORT

INDEPENDENT DELIVERY

MODEL ANSWER

"In HTTP/2, multiple requests share one TCP connection. If a single packet containing Stream A is lost, TCP pauses the entire connection buffer, blocking Stream B and C. HTTP/3 runs over UDP-based QUIC, where each stream is tracked independently. A lost packet only stalls its specific stream."

DEEP DIVE FOLLOW-UP

→ "What is HTTP Server Push?"

A feature in HTTP/2 where the server proactively sends critical resources (like CSS or JS files) to the client's cache before the client parses the HTML and asks for them.

HTTP VERSION PROGRESSION

HTTP/1.1: Persistent connections, pipelining, chunked transfer encoding.

HTTP/2: Binary framing, multiplexing over single TCP connection, server push.

HTTP/3: Replaces TCP with QUIC over UDP, eliminating Head-of-Line blocking.

WHERE USED IN INDUSTRY

Browser engines fetch web documents using HTTP/2 multiplexing, allowing multiple image and script assets to load in parallel over a single TCP stream.

INTERVIEW DEPTH

L1: HTTP Request (GET/POST/PUT) and Response (status codes) structures.

L2: HTTP/1.1 persistent connections, pipelining, and HoL blocking.

L3: HTTP/2 binary framing, stream multiplexing, and server push.

L4: HTTP/3 QUIC stream independence and connection migration.

L5: Browser caching controls (Cache-Control, ETag validation).

COMMON MISTAKE

Believing that textbook definitions are sufficient; always lead with real-world trade-offs.

TOP FOLLOW-UPS

- What is the difference between HTTP/1.1 and HTTP/2?
- Explain HTTP status code classes.

INTERVIEWER TRAP

Don't say HTTP/3 is less secure because it uses UDP. Security (TLS 1.3) is natively embedded into QUIC's transport protocol, unlike TCP where it's a layer on top.

MEMORY TRICK

HTTP/1.1 = One road with toll gates. HTTP/2 = Multi-lane highway, but one crash blocks all lanes. HTTP/3 = Multiple independent monorails.

APPLICATION LAYER PROTOCOLS

Yield: ★★★★★

10 – HTTPS & SSL/TLS Handshake

Encrypting web traffic to prevent eavesdropping and modification.

CRYPTOGRAPHIC ARCHITECTURE

HTTPS = HTTP over SSL/TLS. Uses **Asymmetric Cryptography** (Public/Private keys) to negotiate a temporary, shared key, then uses **Symmetric Cryptography** (AES) for fast session data encryption.

1. Client sends
ClientHello
(Cipher suites, random)
2. Server replies
ServerHello
, Cert, Public Key
3. Client verifies Cert with Root CA database
4. Client encrypts
Pre-Master Secret
with Server's Public Key
5. Both derive matching
Session Key
(Symmetric)

PUBLIC KEY INFRASTRUCTURE (PKI)

Certificates are cryptographically signed by Trusted Certificate Authorities (CAs) like Let's Encrypt. Browsers ship pre-loaded with these CA public keys to verify signatures.

WHERE USED IN INDUSTRY

Banking web applications run TLS handshakes to negotiate symmetric session keys, ensuring all customer transactions remain encrypted and tamper-proof.

INTERVIEW DEPTH

- L1:** HTTPS port 443 vs HTTP port 80 credentials security.
- L2:** TLS handshake phases: Cipher negotiation, Key exchange (DH), Cert validation.
- L3:** Asymmetric encryption (session setup) vs symmetric encryption (payload).
- L4:** Certificate Authorities (CAs), trust store root certs, and CRLs.
- L5:** ALPN negotiation and TLS 1.3 0-RTT handshakes.

CORE INTERVIEW QUESTION

"Why does HTTPS use both **Asymmetric and Symmetric encryption** instead of just **Asymmetric**?"

KEYWORDS TO DROP

COMPUTATIONAL OVERHEAD

KEY EXCHANGE

AES ENCRYPTION

SYMMETRIC SESSION KEY

MODEL ANSWER

"Asymmetric encryption is computationally expensive and slow for encrypting large amounts of data. HTTPS uses asymmetric encryption solely to securely exchange a symmetric session key. Once shared, fast symmetric encryption (like AES) secures the actual web traffic."

DEEP DIVE FOLLOW-UP

→ "What is **Perfect Forward Secrecy (PFS)**?"

A feature of key-agreement protocols (like Diffie-Hellman) that ensures session keys are unique. Compromise of the server's private key does not decrypt past recorded sessions.

REAL INTERVIEW SCENARIO

Interviewer: "Why do we use symmetric encryption in HTTPS if asymmetric is more secure?"

Candidate: "Symmetric encryption isn't less secure; it's just much faster. We use asymmetric encryption solely to safely exchange the symmetric session key, which encrypts payload."

COMMON MISTAKE

Believing that textbook definitions are sufficient; always lead with real-world trade-offs.

TOP FOLLOW-UPS

- Why is symmetric encryption used for payload transit?
- What is SNI in TLS?

INTERVIEWER TRAP

Don't say HTTPS encrypts the IP headers. It only encrypts the HTTP payload, parameters, headers, and cookies. Attackers can still see the destination IP address.

MEMORY TRICK

Asymmetric = Safe box key delivery. Symmetric = The actual padlock coding used for bulk transport.

TRANSPORT LAYER PROTOCOLS

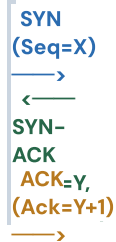
Yield: ★★★★★

11 – TCP 3-Way Handshake

How TCP establishes a reliable bidirectional connection.

CONNECTION SYNCHRONIZATION

Ensures that both client and server are ready for transmission, synchronizes sequence numbers, and allocates buffer space.



VULNERABILITY: SYN FLOOD ATTACK

An attacker sends thousands of SYN packets but never sends the final ACK. The server leaves connections "half-open" (SYN_RCVD), exhausting system memory buffers.

WHERE USED IN INDUSTRY

Web servers scale their backlog queues to manage concurrent incoming TCP SYN packets, preventing half-open connection memory exhaustion.

INTERVIEW DEPTH

- L1:** Three-way handshake states: SYN, SYN-ACK, ACK.
- L2:** Initial Sequence Number (ISN) generation logic.
- L3:** Half-open queue (SYN backlog) and accept queue allocations.
- L4:** SYN flood attacks and cookie-based stateless defenses.
- L5:** TCP Fast Open (TFO) payload exchange during handshake.

CORE INTERVIEW QUESTION

"Why is the TCP handshake 3 steps instead of 2?"

KEYWORDS TO DROP

BIDIRECTIONAL SYNC

HALF-OPEN CONNECTION

SEQUENCE NUMBERS

TWO-WAY CONFIRMATION

MODEL ANSWER

"A reliable connection requires both sides to verify that they can transmit AND receive data. Step 1 verifies Client-to-Server path. Step 2 verifies Server-to-Client path. Step 3 confirms to the server that the client can hear it. 2 steps would leave the server uncertain of the client's receipt."

DEEP DIVE FOLLOW-UP

→ "What is a SYN Cookie?"

A defense mechanism where the server doesn't allocate buffer memory immediately. It encodes session information into the SYN-ACK sequence number (cookie) and verifies it upon final ACK.

REAL INTERVIEW SCENARIO

Interviewer: "Why isn't a 2-way handshake enough to establish a TCP connection?"

Candidate: "A 2-way handshake only confirms the client can reach the server. The client cannot verify if the server received its acknowledgment, leaving the connection half-open."

COMMON MISTAKE

Believing that textbook definitions are sufficient; always lead with real-world trade-offs.

TOP FOLLOW-UPS

- Why is a 2-way handshake insufficient?
- What is a SYN flood attack?

INTERVIEWER TRAP

Don't forget that data CAN be sent inside the final ACK packet of the handshake, although traditional implementations wait.

MEMORY TRICK

Handshake = 1. 'Hello?' 2. 'I hear you, do you hear me?' 3. 'Yes, I hear you!'

TRANSPORT LAYER PROTOCOLS

Yield: ★★★★★

12 – TCP Connection Termination

The 4-way termination handshake to gracefully close active sessions.

HALF-CLOSE STATE

TCP connection is full-duplex. Closing the connection is done independently for each direction, hence requiring a 4-step wave.



WHY THE TIME_WAIT STATE EXISTS

The host that initiates the active close enters a `TIME_WAIT` state for $2 \times \text{MSL}$ (Maximum Segment Lifetime). This ensures the final ACK arrived safely at the server and clears stray packets.

WHERE USED IN INDUSTRY

High-performance proxy servers (like Nginx) monitor sockets in `TIME_WAIT` state to ensure late arriving packet fragments don't pollute new connections.

INTERVIEW DEPTH

- L1:** Four-way handshake states: FIN, ACK, FIN, ACK.
- L2:** Half-closed states (`FIN_WAIT_1`, `FIN_WAIT_2`, `CLOSE_WAIT`).
- L3:** `TIME_WAIT` state purpose ($2 \times \text{MSL}$ wait duration).
- L4:** Socket reuse problems and `SO_REUSEADDR` socket options.
- L5:** RST (Reset) flag overrides for immediate termination.

CORE INTERVIEW QUESTION

"Why does TCP require 4 steps to terminate a connection but only 3 steps to establish it?"

KEYWORDS TO DROP

INDEPENDENT CLOSE

HALF-CLOSED STATE

FULL-DUPLEX CHANNEL

FIN PACKET

MODEL ANSWER

"Because TCP is full-duplex, closing one direction doesn't mean the other side is ready to stop sending. The client sends FIN. The server ACKs to close Client→Server traffic. Only when the server completes its pending transmissions will it send its own FIN. The client then ACKs to close Server→Client."

DEEP DIVE FOLLOW-UP

→ "What is a TCP Reset (RST) packet?"

A hard termination indicator sent when a host receives a packet for a port that isn't open, or during client-side crashes where a connection is immediately aborted without a handshake.

TEARDOWN STATE PROGRESSION

- Level 1 (FIN):** Host A sends FIN segment indicating it has no more data.
- Level 2 (ACK):** Host B acknowledges FIN, entering `CLOSE_WAIT` state.
- Level 3 (FIN-2):** Host B finishes sending its data, then sends its own FIN.
- Level 4 (TIME_WAIT):** Host A enters `TIME_WAIT` ($2 \times \text{MSL}$) to ensure final ACK is received.

COMMON MISTAKE

Believing that textbook definitions are sufficient; always lead with real-world trade-offs.

TOP FOLLOW-UPS

- Explain the 4-way TCP teardown sequence.
- Why is `TIME_WAIT` state necessary?

INTERVIEWER TRAP

Don't confuse `CLOSE_WAIT` with `TIME_WAIT`. `CLOSE_WAIT` is a passive close state indicating the application code must release the socket resources.

MEMORY TRICK

FIN: I am done. ACK: Got it, let me finish. FIN: I am also done. ACK: Done.

TRANSPORT LAYER MECHANISMS

Yield: ★★★★★

13 – Flow Control

Preventing fast senders from overwhelming slow receivers.

RECEIVER BUFFER MANAGEMENT

Flow control matches the sender's transmission rate to the receiver's processing capabilities. Controlled entirely by the receiver's available buffer space.

TCP Header: Window Size

[Sender] — Data (Seq=1, Size=1000B) [Receiver]
[Sender] <— ACK (Ack=1001, Win=4000B) [Receiver]

WINDOW STARVATION & PROBE SEGMENT

If receiver window drops to 0, sender stops. To prevent deadlock, the sender periodically transmits 1-byte **Zero Window Probes** to poll if buffer has cleared.

WHERE USED IN INDUSTRY

Data processing pipelines tune the TCP receive window size to slow down high-speed senders when the local buffer memory runs full.

INTERVIEW DEPTH

- L1: End-to-end flow control vs network congestion control.
- L2: TCP Advertised Window field in header blocks.
- L3: Sender sliding window adjustments based on receiver buffers.
- L4: Zero Window conditions and keep-alive probe packets.
- L5: Silly Window Syndrome prevention using Nagle's algorithm.

CORE INTERVIEW QUESTION

"What mechanism does TCP use to advertise receiver buffer availability dynamically?"

KEYWORDS TO DROP

SLIDING WINDOW

ADVERTISED WINDOW

TCP HEADER FIELD

ZERO WINDOW PROBE

MODEL ANSWER

"TCP uses the Sliding Window protocol. The receiver advertises its available buffer space in the 16-bit Window Size field of every ACK packet. The sender calculates: $\text{Max Sent} = \text{Last Byte Acked} + \text{Advertised Window}$. This dynamically throttles the sender."

DEEP DIVE FOLLOW-UP

→ "What is the Window Scale option?"

Since the 16-bit window field maxes out at 64KB, the TCP option 'Window Scale' multiplies this value exponentially to support high-throughput, high-latency links.

FLOW CONTROL PROGRESSION

Level 1 (Scope): Flow control is end-to-end (protects receiver buffer overflow).

Level 2 (Field): Controlled by the 16-bit Window Size field in the TCP header.

Level 3 (Scaling): Window Scaling option allows windows up to 1GB.

Level 4 (Zero Window): Sender stops sending when window is 0; polls with keep-alives.

COMMON MISTAKE

Believing that textbook definitions are sufficient; always lead with real-world trade-offs.

TOP FOLLOW-UPS

- What is the window size field in TCP header?
- What is sliding window window scaling?

INTERVIEWER TRAP

Don't confuse Flow Control (matching sender speed to receiver speed) with Congestion Control (matching sender speed to network capacity).

MEMORY TRICK

Flow Control is the receiver saying 'Don't talk too fast, my ears are full!'

TRANSPORT LAYER MECHANISMS

Yield: ★★★★★

14 – Sliding Window Protocol

Enabling efficient continuous transmission without waiting for individual ACKs.

PIPELINED TRANSMISSION

Instead of Stop-and-Wait (sending one packet and blocking until it is acknowledged), the sliding window allows sending multiple unacknowledged packets to maximize throughput.

```
[Aked] [ Sent but Unacked | Can Send ] [Cannot
        Send]
Bytes:  1 2 3 [ 4 5 6 7 | 8 9 10 ] 11 12
        ← Window Size = 7 Bytes →
```

GO-BACK-N VS SELECTIVE REPEAT

Go-Back-N (GBN): Simplistic. If packet 3 is lost, receiver discards 4 and 5, forcing sender to retransmit everything starting from packet 3.

Selective Repeat (SR): Efficient. Receiver buffers out-of-order packets (4, 5) and ACKs them; sender only retransmits lost packet 3.

WHERE USED IN INDUSTRY

File transfer protocols (like SFTP) utilize sliding window sequences to transmit batch segments before receiving individual acknowledgments.

INTERVIEW DEPTH

- L1:** Sliding window boundaries (Send window, Receive window).
- L2:** Stop-and-Wait protocol inefficiency.
- L3:** Go-Back-N (GBN) retransmission rules on packet losses.
- L4:** Selective Repeat (SR) individual buffering and dedicated ACKs.
- L5:** Bandwidth-Delay Product (BDP) capacity scaling.

CORE INTERVIEW QUESTION

"How does TCP implement pipelined delivery, and what determines its window boundaries?"

KEYWORDS TO DROP

CUMULATIVE ACKS

GO-BACK-N

OUT-OF-ORDER BUFFER

PIPELINING

MODEL ANSWER

"TCP implements sliding window using cumulative ACKs. The window boundaries are dictated by sequence numbers. The left boundary slides forward when the lowest unacknowledged sequence number receives an ACK, allowing new packets to fall inside the window on the right."

DEEP DIVE FOLLOW-UP

→ "What is Selective ACK (SACK) in TCP?"

A TCP extension allowing receivers to explicitly advertise non-contiguous ranges of successfully received segments, enabling selective retransmissions.

WINDOW PROTOCOL COMPARISON

Stop & Wait: Sender sends 1 packet and waits for ACK before sending next.

Go-Back-N: Sends N packets. On loss, retransmits all packets starting from loss.

Selective Repeat: Sends N packets. Receiver buffers out-of-order; retransmits only loss.

COMMON MISTAKE

Believing that textbook definitions are sufficient; always lead with real-world trade-offs.

TOP FOLLOW-UPS

- Contrast Go-Back-N vs Selective Repeat.
- What is receiver buffer exhaustion?

INTERVIEWER TRAP

Don't say TCP is strictly Go-Back-N. Standard modern TCP implementations use SACK (Selective Acknowledgment) which functions like Selective Repeat.

MEMORY TRICK

Sliding window is like a moving frame over a conveyor belt. Only what's inside the frame can be processed.

TRANSPORT LAYER MECHANISMS

Yield: ★★★★★

15 – TCP Congestion Control

Avoiding network collapse by dynamically adjusting transmission windows.

SENDER-SIDE NETWORK GUARD

Protects routing queues from overflow. Governed by `cwnd` (Congestion Window) which adjusts based on network congestion indicators.

Phase	Trigger / Action	cwnd Behavior
Slow Start	Initial phase	Doubles every RTT
Congestion Avoidance	$cwnd > ssthresh$	Increments by 1 MSS / RTT
Fast Recovery	3 Duplicate ACKs	Halves <code>cwnd</code> , sets <code>ssthresh</code>

CONGESTION NOTIFICATION (ECN)

Rather than dropping packets, modern routers mark the ECN bits in the IP header of packets when buffers are full. Receivers echo this to senders to preemptively reduce rates.

WHERE USED IN INDUSTRY

Video streaming services (like Netflix) throttle data rates using TCP congestion windows (AIMD algorithm) when network packet drops indicate path overload.

INTERVIEW DEPTH

- L1: Congestion triggers (buffer bloat, packet drops, high delays).
- L2: Slow Start exponential `cwnd` expansion up to `ssthresh`.
- L3: Congestion Avoidance linear AIMD growth rules.
- L4: Fast Retransmit (3 duplicate ACKs) and Fast Recovery.
- L5: Congestion algorithms comparisons (Tahoe, Reno, Cubic, BBR).

CORE INTERVIEW QUESTION

"Explain the difference in how TCP reacts to a Packet Timeout vs receiving 3 Duplicate ACKs."

KEYWORDS TO DROP

SEVERE CONGESTION

FAST RETRANSMIT

SSTHRESH HALVED

CWND RESET

MODEL ANSWER

"A Timeout indicates severe congestion; TCP resets `cwnd` to 1 MSS, sets `ssthresh` to half of current `cwnd`, and returns to Slow Start. 3 Duplicate ACKs trigger Fast Retransmit: TCP halving `cwnd` (Multiplicative Decrease) and entering Congestion Avoidance, assuming some packets are still passing."

DEEP DIVE FOLLOW-UP

→ "What is AIMD?"

Additive Increase Multiplicative Decrease. TCP increases rate slowly (linear) to test capacity, but drops rate aggressively (half) if congestion is detected to ensure stability.

CONGESTION PHASES

Slow Start: Exponential `cwnd` growth (doubles every RTT) up to `ssthresh`.

Congestion Avoidance: Linear `cwnd` growth (adds 1 MSS per RTT) via AIMD.

Fast Retransmit: Retransmits packet on receiving 3 duplicate ACKs.

Fast Recovery: Drops `cwnd` to half of current size, skips slow start back to linear.

COMMON MISTAKE

Believing that textbook definitions are sufficient; always lead with real-world trade-offs.

TOP FOLLOW-UPS

- Explain TCP Slow Start phase.
- How does AIMD work in congestion avoidance?

INTERVIEWER TRAP

Don't confuse `ssthresh` (Slow Start Threshold) with `cwnd`. `ssthresh` marks the switchover point from exponential growth to linear growth.

MEMORY TRICK

Timeout = Panic button (reset to 1). 3 Dup ACKs = Caution signal (cut speed in half).

DATA LINK LAYER & ROUTING

Yield: ★★★★★

16 – MAC vs IP Addressing

Physical hardware addressing vs logical network location addressing.

KEY IDENTIFIERS

Physical MAC Address: Flat, non-hierarchical space. Assigned by manufacturer. Burned into the Network Interface Card (NIC). Relates to physical link layer.

Logical IP Address: Hierarchical network addressing. Assigned dynamically. Relates to logical location and routing capabilities.

MAC Address Structure (48-bit)

[3 Octets OUI] [3 Octets NIC serial]
e.g., 00:1A:2B : 3C:4D:5E

IP Address Structure (32-bit)

[Network ID] [Host ID]

WHERE USED IN INDUSTRY

Local area networks implement MAC address filtering on switches to block unregistered devices from tapping corporate Ethernet ports.

INTERVIEW DEPTH

- L1:** 48-bit hex layout (OUI vs NIC specific blocks).
- L2:** Unicast, Multicast, and Broadcast MAC targets.
- L3:** How switches learn MAC addresses in CAM tables.
- L4:** MAC flooding attacks and port security limits.
- L5:** MAC address randomization protocols on mobile devices.

CORE INTERVIEW QUESTION

"Why do we need both MAC and IP addresses? Why can't we use MAC addresses globally?"

KEYWORDS TO DROP

HIERARCHICAL ROUTING

GLOBAL TABLE SIZE

HARDWARE INDEPENDENCE

FLAT NAMESPACE

MODEL ANSWER

"MAC addresses belong to a flat namespace with no geographical or routing context. If we routed using MACs, every router on Earth would need a routing table entry for billions of individual devices, causing memory collapse. Logical IPs segment hosts into network blocks, enabling hierarchical routing."

DEEP DIVE FOLLOW-UP

→ "Can a MAC address be spoofed?"

Yes. Although burned into hardware, operating systems allow software overriding of the MAC address field in outbound Ethernet headers.

REAL INTERVIEW SCENARIO

Interviewer: "What happens to the MAC addresses of a frame when passing a router?"

Candidate: "The router strips the source MAC and sets it to its egress port MAC. It sets the destination MAC to the next hop's ingress port MAC. IP addresses remain constant."

COMMON MISTAKE

Believing that textbook definitions are sufficient; always lead with real-world trade-offs.

TOP FOLLOW-UPS

- How does a switch build its MAC table?
- Contrast L2 switch forwarding vs L3 routing.

INTERVIEWER TRAP

Don't say that MAC addresses are used in routers. Routers strip the MAC header entirely, inspect the IP header, and append a new MAC header before egress.

MEMORY TRICK

MAC is your SSN (fixed identifier). IP is your mailing address (changes when you move).

NETWORK LAYER CONCEPTS

Yield: ★★★★★

17 – Routing vs Forwarding

Determining network pathways vs switching packets to exit interfaces.

PLANE SEPARATION

Routing (Control Plane): Process of determining the end-to-end path a packet should take. Slow, calculated in software.

Forwarding (Data Plane): Act of transferring a packet from a router's input port to the appropriate output port. High-speed, executed in hardware.

Control Plane (Routing Table)

↓ populates

Data Plane (Forwarding Table / FIB)

WHERE USED IN INDUSTRY

Core Internet routers inspect global BGP routing tables to forward IP packets along the shortest path toward their destination subnet.

INTERVIEW DEPTH

- L1:** Routing Tables: Destination prefix, next hop, egress interface.
- L2:** Longest Prefix Match (LPM) algorithm in hardware lookup tables.
- L3:** Static routing vs default routing (0.0.0.0/0).
- L4:** Administrative Distance (AD) and metric comparisons.
- L5:** IP Time-to-Live (TTL) decrementing and ICMP Time Exceeded packets.

CORE INTERVIEW QUESTION

"How do routing tables differ from forwarding tables inside a router?"

KEYWORDS TO DROP

ROUTING TABLE (RIB)

FORWARDING TABLE (FIB)

ASIC HARDWARE

NEXT-HOP RESOLUTION

MODEL ANSWER

"The Routing Table (RIB) is a control-plane database populated by routing protocols (BGP, OSPF) containing all potential paths. The Forwarding Table (FIB) is a streamlined data-plane table loaded directly into ASIC hardware, optimized for ultra-fast next-hop lookup for active packets."

DEEP DIVE FOLLOW-UP

→ "What is policy-based routing?"

A technique where routing decisions are made based on metrics beyond destination IP (such as source IP, port numbers, or protocol types).

ROUTING DEPTH PROGRESSION

- Level 1 (Logic):** Router routes IP packets between subnets using a routing table.
- Level 2 (Static):** Administrator manually configures routes (used in stub networks).
- Level 3 (Dynamic):** Routing protocols dynamically exchange and calculate paths.
- Level 4 (Tables):** Inspecting destination IP prefix matches for longest prefix match.

COMMON MISTAKE

Routing based on MAC addresses instead of IPs.

TOP FOLLOW-UPS

- What is autonomous system routing?
- Contrast static routing vs dynamic routing.

INTERVIEWER TRAP

Don't confuse the two. If an interviewer asks how a router processes packets at 100Gbps, the answer is 'Forwarding table hardware (ASIC)', not routing calculations.

MEMORY TRICK

Routing is planning the route on a map (Control). Forwarding is turning the steering wheel at the intersection (Data).

NETWORK LAYER ROUTING

Yield: ★★★★★

18 – Routing Protocols (OSPF vs BGP)

Interior gateway routing (IGP) vs exterior gateway routing (EGP).

SCALE & SCOPE

OSPF (Open Shortest Path First): IGP link-state protocol. Used within an Autonomous System (AS). Metric is 'Cost' (bandwidth). Fast convergence.

BGP (Border Gateway Protocol): EGP path-vector protocol. Used to route traffic between Autonomous Systems. Metrics are policy-based attributes.

OSPF (Link State)
Dijkstra Algorithm
Metric: Cost
Updates: Multicast

BGP (Path Vector)
Path Vector
Metric: Policies
Updates: TCP Port 179

 WHERE USED IN INDUSTRY

Corporate datacenters run Open Shortest Path First (OSPF) routing protocols to automatically recalculate paths when a fiber optic link breaks.

 INTERVIEW DEPTH

- L1:** Interior Gateway Protocols (IGP) vs Exterior Gateway Protocols (EGP).
- L2:** Distance Vector (RIP) hop count loops and split horizon rules.
- L3:** Link State (OSPF) Dijkstra calculation and link state databases.
- L4:** Path Vector (BGP) policy-based routing and TCP port 179 sessions.
- L5:** Fast convergence algorithms in routing fabrics.

CORE INTERVIEW QUESTION

"Why does the Internet run on BGP rather than OSPF?"

KEYWORDS TO DROP

AUTONOMOUS SYSTEMS

POLICY ROUTING

DIJKSTRA SCALE LIMIT

ROUTE SUMMARIZATION

MODEL ANSWER

"OSPF requires all routers to compute Dijkstra's algorithm to map the entire network topology, which is computationally impossible at internet scale. BGP connects independent Autonomous Systems (AS) and routes based on business policies and paths (AS-Path), not raw bandwidth metrics."

DEEP DIVE FOLLOW-UP

→ **"What is the difference between iBGP and eBGP?"**

eBGP routes traffic between different Autonomous Systems. iBGP distributes those external paths to internal routers within the same Autonomous System.

 IGP VS EGP COMPARISON

OSPF (IGP Link-State): Uses Dijkstra. Fast convergence. Operates inside Autonomous Systems.

BGP (EGP Path-Vector): Uses TCP port 179. Policy-based routing. Connects different Autonomous Systems.

 COMMON MISTAKE

Believing that textbook definitions are sufficient; always lead with real-world trade-offs.

 TOP FOLLOW-UPS

- How does OSPF exchange routing updates?
- Why does BGP use TCP port 179?

 INTERVIEWER TRAP

Don't confuse RIP with modern routing. RIP is a distance-vector protocol limited to 15 hops; it is rarely used in modern production networks.

 MEMORY TRICK

OSPF is a turn-by-turn city GPS map. BGP is a map showing interstate highway systems between major shipping ports.

NETWORK LAYER SERVICES

Yield: ★★★★★

19 – Network Address Translation (NAT)

Conserving IPv4 addresses by sharing public IPs.

PRIVATE ADDRESS TRANSLATION

Private IP ranges (e.g. 192.168.x.x, 10.x.x.x) are non-routable on the internet. A NAT router translates these private addresses to a single public IP address.

Local Host (192.168.1.10:8000)
↓ packets egress
NAT Router Translates to:
Public IP (203.0.113.1:54320)
(Maintains mapping in translation table)

PORT ADDRESS TRANSLATION (PAT)

The most common form of NAT (also called NAT Overload). Maps thousands of internal hosts to a single public IP by tracking distinct source port numbers.

WHERE USED IN INDUSTRY

Home routers use Port Address Translation (PAT) to share a single public IP address among dozens of smart devices, conserving global IPv4 address space.

INTERVIEW DEPTH

- L1:** Static NAT, Dynamic NAT, and Port Address Translation (PAT).
- L2:** NAT translation tables (Private IP + Port $\leftrightarrow$ Public IP + Port).
- L3:** NAT traversal techniques (STUN, TURN) for P2P connections.
- L4:** Hairpinning NAT local loopbacks.
- L5:** Double NAT routing issues and IPv6 native address paths.

CORE INTERVIEW QUESTION

"How does a NAT router know which internal host should receive an incoming packet?"

TRANSLATION TABLE

NAT OVERLOAD / PAT

SOURCE PORT ALLOCATION

STATEFUL MAPPING

MODEL ANSWER

"The NAT router maintains a dynamic NAT Translation Table. When an internal host sends a packet, the router rewrites the source IP and source port, recording this mapping. When a response returns, the router references the destination port against this table to identify the host."

DEEP DIVE FOLLOW-UP

→ **"What is Static NAT?"**

A 1-to-1 mapping where a specific internal server is assigned a dedicated public IP, allowing incoming external connections (e.g. hosting a web server).

REAL INTERVIEW SCENARIO

Interviewer: "Why do we need Port Address Translation (PAT) instead of basic NAT?"

Candidate: "Basic NAT requires a 1:1 mapping of private IPs to public IPs. PAT maps multiple private IPs to a single public IP by translating port numbers, saving public IPs."

COMMON MISTAKE

Thinking NAT is a security firewall. It just hides internal IPs.

TOP FOLLOW-UPS

- Differentiate Static NAT, Dynamic NAT, and PAT.
- What are the security limits of NAT?

INTERVIEWER TRAP

Don't say NAT is a security tool. While it blocks unsolicited incoming requests by default, its primary design goal was conserving IPv4 addresses.

MEMORY TRICK

NAT is like a corporate telephone switchboard: one public number, but calls get routed internally by extension numbers.

NETWORK HARDWARE & INFRASTRUCTURE

Yield: ★★★★★

20 – Network Devices

Analyzing Hubs, Switches, Routers, and L3 Switches.

LAYER SEGMENTATION

Hub (L1): Multi-port repeater. Broadcasts all data to all ports. Creates a single massive collision domain.

Switch (L2): Learns MAC addresses to forward frames selectively, creating isolated collision domains per port.

Router (L3): Uses IP tables to forward packets between networks. Limits broadcast domains.

Device	Layer	Collision Domain	Broadcast Domain
Hub	L1	1 (All ports)	1 (All ports)
Switch	L2	Isolated per port	1 (All ports)
Router	L3	Isolated per port	Isolated per port

WHERE USED IN INDUSTRY

Network engineers deploy Layer 3 switches at core server racks for line-rate packet forwarding, while using hubs only in legacy labs.

INTERVIEW DEPTH

- L1:** Hubs, Switches, Routers, and Gateways.
- L2:** Collision domains vs broadcast domains boundaries.
- L3:** Half-duplex vs full-duplex CSMA/CD logic.
- L4:** Layer 3 switches vs Layer 7 load balancers.
- L5:** Software-Defined Networking (SDN) data plane vs control plane.

CORE INTERVIEW QUESTION

"What happens to collision and broadcast domains when you replace a Hub with a L2 Switch?"

KEYWORDS TO DROP

COLLISION ISOLATION

MAC LEARNING TABLE

BROADCAST DOMAIN WIDTH

MICROSEGMENTATION

MODEL ANSWER

"Replacing a Hub with a Switch breaks one massive collision domain into separate collision domains for each individual port, preventing packets from colliding. The broadcast domain, however, remains unchanged; broadcasts are still forwarded out of all switch ports."

DEEP DIVE FOLLOW-UP

→ **"How does a Switch build its MAC table?"**

By inspecting the source MAC address of incoming frames. If the MAC is not in its table, it associates the port with that address. If a destination is unknown, it floods the frame.

DEVICE DOMAIN PROGRESSION

Hub (L1): Shared collision domain, shared broadcast domain. Half duplex.

Switch (L2): Separate collision domain per port, shared broadcast domain. Full duplex.

Router (L3): Separate collision domain, separate broadcast domain. IP isolation.

COMMON MISTAKE

Believing that textbook definitions are sufficient; always lead with real-world trade-offs.

TOP FOLLOW-UPS

- Explain the collision domains of hub vs switch.
- What is a broadcast storm?

INTERVIEWER TRAP

Don't confuse a L3 switch with a Router. A router handles diverse WAN interface types (fiber, coaxial, serial), while L3 switches primarily route Ethernet connections.

MEMORY TRICK

Hubs yell to everyone. Switches speak to who they know. Routers look up directions to foreign cities.

NETWORK CORE TECHNOLOGIES

Yield: ★★★★★

21 – Circuit vs Packet Switching

Dedicated paths vs dynamically routed independent packets.

FUNDAMENTAL PARADIGMS

Circuit Switching: Establishes a dedicated physical path between sender and receiver for the duration of the call (e.g. traditional telephone system).

Packet Switching: Breaks data into small packet units sent independently across shared nodes (e.g. internet routing).

Circuit Switching
Dedicated bandwidth
No packet overhead
Poor channel utility

Packet Switching
Shared pathways
Packet queues & delay
High channel utility

WHERE USED IN INDUSTRY

Virtual Private Clouds (VPCs) run virtual switches to isolate broadcast traffic between tenants inside physical hypervisor chassis.

INTERVIEW DEPTH

- L1: Store-and-forward vs cut-through vs fragment-free switching.
- L2: VLAN tagging (802.1Q headers) and trunk port trunks.
- L3: Spanning Tree Protocol (STP) loop prevention (blocking states).
- L4: CAM table overflows and switch replication floods.
- L5: Link Aggregation (LACP) bandwidth groupings.

CORE INTERVIEW QUESTION

"Why does the internet run on Packet Switching rather than Circuit Switching?"

KEYWORDS TO DROP

BANDWIDTH UTILIZATION

BURST TRAFFIC

SINGLE POINT OF FAILURE

QUEUEING DELAY

MODEL ANSWER

"Circuit switching is highly inefficient for bursty computer data, as reserved bandwidth sits idle when no data is sent. Packet switching allows thousands of users to share the same physical cable link. If a node fails, packets are dynamically rerouted, avoiding connection loss."

DEEP DIVE FOLLOW-UP

→ **"What is statistical multiplexing?"**

A method where communication channels are shared dynamically based on demand, allocating transmission slots only to hosts actively sending packets.

REAL INTERVIEW SCENARIO

Interviewer: "What is a Layer 3 Switch and when do you use it instead of a router?"

Candidate: "A L3 switch performs routing in hardware using ASICs, which is much faster. It's used for high-speed routing between local VLANs, whereas routers connect WANs."

COMMON MISTAKE

Believing that textbook definitions are sufficient; always lead with real-world trade-offs.

TOP FOLLOW-UPS

- Contrast Layer 2 switches with Layer 3 switches.
- What is a VLAN tag?

INTERVIEWER TRAP

Don't say packet switching is always faster. Circuit switching has zero queueing delay or jitter once established, making it optimal for legacy voice calls.

MEMORY TRICK

Circuit is booking a private train lane. Packet is sending letters via the post office.

NETWORK SECURITY & VPNS

Yield: ★★★★★☆

22 – VPN & Tunneling

Establishing secure encrypted channels over untrusted public networks.

ENCAPSULATION & CRYPTOGRAPHY

A Virtual Private Network (VPN) creates an encrypted tunnel between a client and a gateway. Typically implemented using **IPsec** or **SSL/TLS**.

Client Host [Data | Original IP Header]
↓ Encapsulated by IPsec Tunnel Mode
[New IP Header | **ESP Header (Encrypted Payload)**]

IPSEC: TRANSPORT VS TUNNEL MODE

Transport Mode: Only encrypts the IP payload. Keeps the original IP headers intact. Used end-to-end.

Tunnel Mode: Encrypts the entire original IP packet and wraps it in a new IP header. Used router-to-router.

WHERE USED IN INDUSTRY

Remote hybrid workforces connect to corporate offices using IPsec VPN tunnels to encrypt private intranet traffic over the public Internet.

INTERVIEW DEPTH

- L1:** Tunneling concept (private packet inside public header).
- L2:** Site-to-Site vs Remote Access VPN structures.
- L3:** IPsec framework: AH (authentication) vs ESP (encryption).
- L4:** IKE (Internet Key Exchange) phases and security associations.
- L5:** TLS/SSL VPNs executing over web browser connections.

CORE INTERVIEW QUESTION

"What is the difference between IPsec Tunnel Mode and Transport Mode?"

KEYWORDS TO DROP

ENCAPSULATING SECURITY PAYLOAD

HEADER ENCRYPTION

GATEWAY TO GATEWAY

HOST TO HOST

MODEL ANSWER

"In IPsec Transport Mode, only the payload of the IP packet is encrypted; the original IP header remains visible, leaving source/destination identities public. In Tunnel Mode, the entire original IP packet (header and payload) is encrypted and nested inside a new IP header, concealing internal IPs."

DEEP DIVE FOLLOW-UP

→ **"What protocol does SSL VPN use?"**

Uses standard TLS on Port 443. This is highly popular because Port 443 is open on almost all firewalls globally, preventing connection blocks.

VPN TUNNELING STATES

Level 1 (Protocol): Encapsulates private packets inside public carrier protocols.

Level 2 (Modes): Tunnel Mode (encrypts full packet) vs Transport Mode (encrypts payload).

Level 3 (IPsec): Uses IKE for key exchange, ESP for encryption, AH for authentication.

Level 4 (SSL VPN): Runs over TLS, enabling secure remote access through web browsers.

COMMON MISTAKE

Thinking a VPN hides traffic volume or origin completely.

TOP FOLLOW-UPS

- Contrast tunnel mode vs transport mode in IPsec.
- What is SSL VPN?

INTERVIEWER TRAP

Don't assume a VPN makes you anonymous on the web. It encrypts traffic from your ISP, but the VPN provider can see all DNS and IP destination routes.

MEMORY TRICK

Transport Mode = Putting a letter in an envelope. Tunnel Mode = Putting the entire envelope inside a security parcel.

NETWORK SECURITY & FIREWALLS

Yield: ★★★★★

23 – Firewalls (Stateful vs WAF)

Inspecting network traffic based on rules, connection state, and application logic.

FIREWALL TAXONOMY

Stateless (L3/4): Checks individual packets against Access Control Lists (ACLs). Quick, but misses multi-packet attack patterns.

Stateful (L3/4): Tracks connection states (SYN, ESTABLISHED). Dynamically permits return traffic.

WAF (L7): Inspects HTTP payloads for application-layer attacks (SQLi, XSS).

CORE INTERVIEW QUESTION

"How does a Web Application Firewall (WAF) differ from a traditional Stateful Firewall?"

KEYWORDS TO DROP

DEEP PACKET INSPECTION

LAYER 7 PAYLOAD

SQL INJECTION (SQLI)

TCP CONNECTION STATE

MODEL ANSWER

"A Stateful firewall operates at Layer 3 and 4, tracking TCP session states to confirm valid connections. It permits all traffic on port 443 blindly. A WAF operates at Layer 7, performing deep packet inspection of the HTTP payload to block threats like SQLi or XSS."

DEEP DIVE FOLLOW-UP

→ **"What is a Next-Generation Firewall (NGFW)?"**

A system combining L3/L4 stateful inspection with L7 application control, inline IPS/IDS prevention, and encrypted SSL decryption capabilities.

FIREWALL PROGRESSION

Packet Filter (Stateless): Checks IP, ports, and protocols against ACL tables.

Stateful Inspection: Tracks active TCP handshake state tables, allowing dynamic responses.

Next-Gen / WAF: Deep packet inspection at Layer 7, validating SQL, scripting payloads.

Type	Layer	Inspection Depth
Stateless	L3 / L4	IP, Port, Protocol
Stateful	L3 / L4	Session State Tracker
WAF	L7	HTTP headers, query, body

WHERE USED IN INDUSTRY

Web Application Firewalls (WAFs) run deep packet inspection to block cross-site scripting (XSS) and SQL injection payloads in HTTP requests.

INTERVIEW DEPTH

- L1:** Stateless packet filtering (ACL rules checking).
- L2:** Stateful packet inspection (connection state tracking tables).
- L3:** Application-level gateway (proxy firewalls).
- L4:** Next-Generation Firewalls (NGFW) deep packet inspection.
- L5:** Egress filtering to prevent command and control callouts.

COMMON MISTAKE

Believing that textbook definitions are sufficient; always lead with real-world trade-offs.

TOP FOLLOW-UPS

- Contrast stateful inspection with stateless ACLs.
- What is a Web Application Firewall?

INTERVIEWER TRAP

Don't state that a WAF is a drop-in replacement for a network firewall. Both are needed: network firewalls block network ports, while WAFs protect web servers.

MEMORY TRICK

Stateless = Bouncer checking IDs. Stateful = Bouncer remembering who went out to smoke. WAF = Bouncer reading your backpack contents.

NETWORK ATTACKS & DEFENSES

Yield: ★★★★★

24 – DDoS Attacks & Mitigation

Flooding system pipelines with distributed botnet traffic.

ATTACK VECTORS

Volumetric Attacks: Flooding the bandwidth pipeline (e.g. DNS Amplification, NTP Reflection).

Protocol Attacks: Exploiting connection state limitations (e.g. TCP SYN Flood, Ping of Death).

Application Layer (L7): Spamming complex database queries or API endpoints (e.g. HTTP GET flood).

DNS Amplification Attack Flow

Attacker (Spoofs Target IP)

↓ sends small request to open DNS servers

DNS Servers → send massive responses to spoofed **Target IP**

WHERE USED IN INDUSTRY

Global traffic scrubbers (like AWS Shield) mitigate SYN floods and reflection attacks using distributed edge scrubbing centers.



INTERVIEW DEPTH

L1: DoS vs DDoS distributed attack architectures.

L2: Volumetric attacks (DNS Amplification, NTP Reflection).

L3: Protocol attacks (SYN floods, Ping of Death).

L4: Application layer attacks (HTTP GET/POST floods).

L5: Anycast routing mitigation and CDN edge absorption.

CORE INTERVIEW QUESTION

"How does a DNS Amplification attack work, and how can it be mitigated?"

KEYWORDS TO DROP

IP SPOOFING

REFLECTION ATTACK

ANYCAST ROUTING

RATE LIMITING

MODEL ANSWER

"An attacker spoofs the source IP of DNS queries to match the target's IP, then queries open DNS resolvers. By requesting large DNSSEC records, a tiny 60-byte request triggers a 4000-byte response. This reflects and amplifies traffic, flooding the target interface."

DEEP DIVE FOLLOW-UP

→ **"What is a CDNs role in DDoS mitigation?"**

CDNs (like Cloudflare) use Anycast routing to distribute massive incoming traffic across globally distributed data centers, preventing load spikes at origin.



REAL INTERVIEW SCENARIO

Interviewer: "How do you protect a web server from a SYN flood attack?"

Candidate: "We use SYN Cookies. The server responds with a SYN-ACK containing a cryptographic hash of parameters as the sequence number, allocating no connection state memory until client replies with ACK."



COMMON MISTAKE

Assuming simple IP blocks stop DDoS. Botnets rotatively shift IPs.



INTERVIEWER TRAP

Don't suggest simple IP blocking as a solution. In a distributed attack (DDoS), thousands of rotative IPs are used, making static blacklisting useless.



TOP FOLLOW-UPS

- How does Anycast routing mitigate DDoS?
- Explain SYN cookie defenses.



MEMORY TRICK

DNS Amplification = Calling a pizza shop, spoofing your neighbor's caller ID, and ordering 50 family boxes to their house.



CN Last-Minute Revision Sheet

Top Formulas, Port Mappings, and High-Yield Summaries

⚡ CRITICAL PORT NUMBERS

DNS	Port 53 (UDP/TCP)
HTTP / HTTPS	Port 80 / 443 (TCP)
DHCP Server/Client	Port 67 / 68 (UDP)
BGP / OSPF	Port 179 (TCP) / Prot. 89
SSH / FTP	Port 22 / Port 20,21 (TCP)

📐 FORMULAS & METRICS

- Usable Subnet Host IPs: $2^{(32 - \text{CIDR})} - 2$
- IPv4 Address Size: 32 bits (4 Bytes)
- IPv6 Address Size: 128 bits (16 Bytes)
- Ethernet MAC Address Size: 48 bits (6 Bytes)
- TCP Header Size: 20 bytes min, 60 bytes max
- UDP Header Size: Fixed 8 bytes

💡 TOP 5 INTERVIEW CONCEPTS TO RECALL

1. **TCP Handshake:** Client (SYN) → Server (SYN-ACK) → Client (ACK). Syncs sequence numbers bidirectionally.
2. **IP vs MAC:** IP is logical location routing context (rewritten nowhere). MAC is hop-by-hop physical link ID (rewritten at every router).
3. **HoL Blocking:** Solved in HTTP/3 by replacing TCP with UDP (QUIC), allowing packet losses to affect only one independent stream.
4. **ARP Process:** Broadcasts local ARP requests to map IP addresses to MAC physical card addresses, caching responses locally.
5. **Stateful Firewalls:** Inspect connection tables dynamically, allowing unsolicited response segments through otherwise closed ports.

🎯 QUICK SELF-TEST CHECKLIST

- ☐ Trace a packet's MAC changes ☐ Draw a TCP 3-way handshake ☐ Calculate a /27 subnet size

CN Comparison Cheat Sheet

Quick Reference Contrast Tables for Fresher Interviews

TCP VS UDP

Feature	TCP	UDP
Connection	Connection-oriented	Connectionless
Reliability	Guaranteed (Retransmission)	Best-effort (No ACK)
Header Size	20 – 60 Bytes	8 Bytes
Ordering	Strict in-order delivery	Out-of-order allowed

HTTP VS HTTPS

Feature	HTTP	HTTPS
Encryption	Plaintext (None)	SSL/TLS Encrypted
Port	Port 80	Port 443
Overhead	Low latency (No handshake)	Higher (TLS handshake)
Security	Vulnerable to MitM	Protected from sniffing

IPv4 VS IPv6

Feature	IPv4	IPv6
Address Size	32-bit (4 Bytes)	128-bit (16 Bytes)
Notation	Dotted decimal (e.g. 192.168.1.1)	Hexadecimal colon (e.g. 2001::1)
Header	Variable (20–60 Bytes)	Fixed (40 Bytes)
Security	IPsec optional	IPsec native/mandatory

MAC VS IP

Feature	MAC Address	IP Address
Layer	Layer 2 (Data Link)	Layer 3 (Network)
Assignment	Physical (NIC manufacturer)	Logical (DHCP/Admin)
Scope	Local network delivery	Global internet routing
Transit Change	Rewritten at every router	Unchanged end-to-end

ROUTER VS SWITCH

Feature	Router	Switch
OSI Layer	Layer 3 (Network)	Layer 2 (Data Link)
Address Used	IP Address	MAC Address
Traffic Scope	Connects different subnets	Connects hosts in same subnet
Table Type	Routing Table	MAC Address Table

HUB VS SWITCH

Feature	Hub	Switch
Layer	Layer 1 (Physical)	Layer 2 (Data Link)
Forwarding	Broadcasts to all ports	Unicasts to targeted port
Collision Domain	Single shared domain	Separate per port
Transmission	Half-duplex (slow)	Full-duplex (fast)

FLOW VS CONGESTION CONTROL

Feature	Flow Control	Congestion Control
Scope	End-to-end (Sender to Receiver)	Network-wide (Routers/Links)
Trigger	Receiver buffer full	Transit link capacity overload
Feedback	Receiver advertised window field	Packet loss (timeouts / triple ACKs)
Mechanisms	Sliding Window	Slow Start, AIMD, Fast Recovery

PROXY VS REVERSE PROXY

Feature	Forward Proxy	Reverse Proxy
Protects	Clients (private LAN)	Servers (backend pool)
Visibility	Client aware; Server doesn't see	Client unaware; sees Proxy IP
Use Case	Bypassing blocks, egress filter	Load balancing, SSL termination
Direction	Outbound requests	Inbound requests

Top Expected Interview Questions

Placement Q&As · Created by Pranav Gawai (Part 1)

Q: What happens when you type 'google.com' in your browser and press Enter?

"First, the browser needs the server's IP address, so it checks local caches like the browser, OS, and router cache. If not found, it queries the DNS resolver, which recursively asks Root, TLD, and Authoritative DNS servers to get the IP. Once the IP is resolved, the browser initiates a TCP 3-way handshake to establish a reliable connection. After that, since it's HTTPS, a SSL/TLS handshake occurs to validate the certificate and generate symmetric session keys. The browser then sends an HTTP GET request, the web server processes it and returns an HTTP 200 response with the HTML/CSS files, and finally the browser's rendering engine parses the documents and displays the visual page."

KEYWORDS: DNS RESOLUTION TCP HANDSHAKE SSL/TLS SESSION KEYS HTTP GET BROWSER RENDERING

Follow-Up: How does DNS caching work? What is SNI in TLS?

Common Mistake: Skipping the TCP/TLS handshakes and saying the HTTP request goes out immediately after DNS resolution.

Expected Depth: Must explain the sequence of handshakes. Senior candidates should details browser layout tree generation.

Q: Why does TCP use a 3-way handshake to establish a connection? Why isn't a 2-way handshake sufficient?

"TCP uses a 3-way handshake to synchronize initial sequence numbers bidirectionally and verify that both the client and server can send and receive data before any application payload is sent. The client sends a SYN, the server responds with SYN-ACK, and the client sends a final ACK. A 2-way handshake is insufficient because it doesn't prevent duplicate historical connections. For example, if a client sends a SYN packet that is delayed in the network, the client might timeout and open a connection via a second SYN. If the delayed SYN finally reaches the server later, a 2-way handshake would force the server to establish a second active connection, wasting resources because the server cannot verify if the client is actually alive and listening."

KEYWORDS: BIDIRECTIONAL SYNCHRONIZATION SEQUENCE NUMBERS HALF-OPEN CONNECTION RESOURCE EXHAUSTION

Follow-Up: What is a SYN flood attack? How does TCP close a connection?

Common Mistake: Claiming that the 3-way handshake is strictly for security. It is actually for reliability and state synchronization.

Expected Depth: Understand how SYN cookies protect against SYN floods during handshake stages.

Top Expected Interview Questions

Placement Q&As · Created by Pranav Gawai (Part 2)

Q: As an IP packet travels from a source laptop to a web server across multiple routers, how do its MAC addresses and IP addresses change?

"The source and destination IP addresses remain completely constant and unchanged end-to-end, since they represent the global logical source and destination of the packet. However, the source and destination MAC addresses are rewritten hop-by-hop at every single router interface. When the packet moves from one physical link to another, the router decapsulates the Layer 2 Ethernet frame, reads the Layer 3 IP header to make a routing decision, and then re-encapsulates the packet into a new frame with the router's egress port MAC as the source and the next-hop router's ingress MAC as the destination."

KEYWORDS: END-TO-END CONSTANT HOP-BY-HOP REWRITE FRAME DECAPSULATION MAC ROUTING BOUNDARY

Follow-Up: What is ARP spoofing? What happens if a router has no route for a destination IP?

Common Mistake: Saying that IP addresses are rewritten at each hop. They are only rewritten if NAT is active at a boundary.

Expected Depth: Differentiate Layer 2 local switching from Layer 3 routed network domains.

Q: What is Head-of-Line (HoL) blocking in TCP, and how does HTTP/3 solve it using QUIC over UDP?

"In TCP, data is delivered as a single ordered byte stream. In HTTP/2, we multiplexed multiple request streams over that single TCP connection. However, if one TCP packet is dropped in transit, TCP pauses the entire connection, halting all streams until that lost packet is retransmitted. This is Head-of-Line blocking. HTTP/3 solves this by replacing TCP with QUIC, which runs on UDP. QUIC implements packet recovery and connection state in user space and treats each stream as completely independent. If a packet on Stream A is dropped, QUIC continues to deliver packets for Stream B and C to the browser immediately, completely eliminating connection-level HoL blocking."

KEYWORDS: MULTIPLEXED STREAMS ORDERED BYTE STREAM QUIC UDP FOUNDATION STREAM INDEPENDENCE CONNECTION MIGRATION

Follow-Up: Why does QUIC run in user space instead of kernel space? What is O-RTT connection setup?

Common Mistake: Saying HTTP/3 is unreliable because it runs over UDP. QUIC handles packet recovery and congestion control, not UDP.

Expected Depth: Explain how connection migration uses connection IDs rather than IP/Ports to maintain sessions during network shifts.

CN Rapid Fire Questions

Fast-Recall Flashcards for Last-Minute Self-Testing

Q: What port does DNS use?

A: Port 53 (UDP for queries, TCP for large responses).

Q: What is the size of an IPv6 address?

A: 128 bits (16 bytes), written in 8 hex groups.

Q: Which protocol runs traceroute?

A: ICMP (via TTL decrementing at each hop).

Q: What is the DORA process in DHCP?

A: Discover, Offer, Request, Acknowledge.

Q: What is loopback for IPv4/IPv6?

A: IPv4: 127.0.0.1; IPv6: ::1 (localhost).

Q: What port does HTTPS run on?

A: Port 443 (HTTP runs on Port 80).

Q: Usable hosts in a /28 subnet?

A: $2^{(32-28)} - 2 = 16 - 2 = 14$ usable host IPs.

Q: What is the Transport Layer PDU?

A: Segment (for TCP) or Datagram (for UDP).

Q: Size of an Ethernet MAC address?

A: 48 bits (6 bytes), physical hardware identifier.

Q: What port does BGP use?

A: TCP Port 179 (OSPF uses IP Protocol 89).

Q: What is PAT in network routing?

A: Port Address Translation; maps many private IPs to one public IP via ports.

Q: What is a default gateway?

A: The router port IP that routes local subnet traffic to external networks.

CN Common Traps & Deflections

Tactics to Evade Trick Interview Questions with Confident Answers

Trap 1: The UDP 'Insecure' Accusation

Interviewer: "Since UDP doesn't establish a connection and has no handshakes, does that mean it is unsafe and insecure to use?"

Deflection: Explain that 'reliability' and 'security' are separate concerns. UDP is connectionless and unreliable (no delivery checks), but security (encryption/auth) is handled at the application layer (e.g., using DTLS or custom payload encryption). UDP is optimal for gaming and VoIP where speed is paramount.

Trap 2: The HTTPS Header Encryption Illusion

Interviewer: "Does HTTPS encrypt the source and destination IP addresses to protect user identity?"

Deflection: State clearly that HTTPS (via TLS) only encrypts the Layer 7 Application payload and headers (like URLs and cookies). The Layer 3 IP header and Layer 4 TCP header must remain in plaintext. If they were encrypted, intermediate routers could not route the packet and firewalls could not track ports.

Trap 3: The DNS Protocol Exclusivity

Interviewer: "Is it true that DNS queries run exclusively on UDP?"

Deflection: Clarify that DNS uses UDP Port 53 for standard name queries because it is fast and lightweight. However, DNS falls back to TCP Port 53 if the response payload exceeds 512 bytes, during zone transfers between primary and secondary nameservers, or when using DNSSEC.

Trap 4: Public Routing of RFC 1918 Private IPs

Interviewer: "What happens if a router on the public internet receives a packet with a destination IP of 192.168.1.50?"

Deflection: State that public internet routers are configured by standard policies to drop RFC 1918 private addresses immediately. Private IPs are non-routable on the public WAN and must go through Network Address Translation (NAT) to map to a valid public IP before leaving the local gateway.

Trap 5: The Handshake Step Redundancy

Interviewer: "Why don't we use a 4-way handshake to establish a TCP connection for extra validation?"

Deflection: Explain that a 3-way handshake is the mathematical minimum required to synchronize initial sequence numbers (ISNs) bidirectionally. A 4th packet would simply be a redundant ACK, adding network round-trip latency without providing any new connection state verification.



Personal Notes & Scribbles